# Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

**Sec:** Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2022-P2-Model)** **Date: 06-05-2026**

**Time: 02.30Pm to 05.30Pm** **GTA-30** **Max. Marks: 180**

# KEY SHEET

## MATHEMATICS

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **6** | 2 | **9** | 3 | **5** | 4 | **6** | 5 | **3** | 6 | **4** |
| 7 | **9** | 8 | **7** | 9 | **AD** | 10 | **A** | 11 | **BCD** | 12 | **AB** |
| 13 | **ABC** | 14 | **ABD** | 15 | **C** | 16 | **D** | 17 | **A** | 18 | **B** |

## PHYSICS

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 19 | **9** | 20 | **6** | 21 | **6** | 22 | **0** | 23 | **3** | 24 | **0** |
| 25 | **5** | 26 | **3** | 27 | **ABD** | 28 | **AD** | 29 | **ABD** | 30 | **BD** |
| 31 | **ACD** | 32 | **ACD** | 33 | **D** | 34 | **C** | 35 | **A** | 36 | **D** |

## CHEMISTRY

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 37 | **1** | 38 | **8** | 39 | **3** | 40 | **5** | 41 | **1** | 42 | **5** |
| 43 | **9** | 44 | **4** | 45 | **AB** | 46 | **ABCD** | 47 | **ABCD** | 48 | **ABCD** |
| 49 | **BC** | 50 | **B** | 51 | **D** | 52 | **A** | 53 | **B** | 54 | **C** |



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# SOLUTIONS
# MATHEMATICS

1. Conceptual

2. $f\left(\cos\frac{\pi}{7}\right)=\sin\frac{\pi}{7}\sin\frac{3\pi}{7}+\sin\frac{3\pi}{7}\sin\frac{5\pi}{7}+\sin\frac{5\pi}{7}\sin\frac{\pi}{7}$

$=\left(\sin\frac{\pi}{7}\right)\left\{\sin\frac{3\pi}{7}+\sin\frac{5\pi}{7}\right\}+\sin\frac{3\pi}{7}\sin\frac{5\pi}{7}$

$=2\sin\frac{\pi}{7}\sin\frac{4\pi}{7}\cos\frac{\pi}{7}+\sin\frac{3\pi}{7}\sin\frac{5\pi}{7}$

$=\sin\frac{2\pi}{7}\sin\frac{4\pi}{7}+\sin\frac{3\pi}{7}\sin\frac{5\pi}{7}$

$=\sin\frac{2\pi}{7}\sin\frac{3\pi}{7}+\sin\frac{3\pi}{7}\sin\frac{5\pi}{7}$ as $\frac{4\pi}{7}=\pi-\frac{3\pi}{7}$

$=\sin\frac{3\pi}{7}\left\{\sin\frac{2\pi}{7}+\sin\frac{5\pi}{7}\right\}$

$=\sin\frac{3\pi}{7}\left\{\sin\frac{2\pi}{7}+\sin\left(\pi-\frac{2\pi}{7}\right)\right\}$

$=\sin\frac{3\pi}{7}\left\{2\sin\frac{2\pi}{7}\right\}=2\sin\frac{2\pi}{7}\sin\frac{3\pi}{7}$

$=\cos\left(\frac{3\pi}{7}-\frac{2\pi}{7}\right)-\cos\left(\frac{2\pi}{7}+\frac{3\pi}{7}\right)$

$=\cos\frac{\pi}{7}-\cos\frac{5\pi}{7}\qquad=\cos\frac{2\pi}{7}+\cos\frac{\pi}{7}$

$f\left(\cos\frac{\pi}{7}\right)=2\cos^2\frac{\pi}{7}+\cos\frac{\pi}{7}-1$

$f\left(\cos\frac{\pi}{7}\right)=a\left(\cos\frac{\pi}{7}\right)^2+b\left(\cos\frac{\pi}{7}\right)+C$

$\therefore a=2,\ b=1,c=-1\qquad\therefore f(x)=2x^2+x-1$

$f(2)=2(2)^2+2-1=9$

3. More generally, let the lengths of the major and minor axes of the ellipse be $2a$ and $2b$, where $a>b$. Place the ellipse in the coordinate plane so that its major axis lies along the $x$-axis, and its minor axis lies along the $y$-axis. Then the ellipse has the equation $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, and its foci are at $(c,0)$ and $(-c,0)$ where $c^2=a^2-b^2$. The parabola with

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vertex at $(0,-b)$ that passes through the foci has equation $y=\frac{b}{c^2}\left(x^2-c^2\right)$. Substituting this expression for $y$ into the equation for the ellipse yields $\frac{x^2}{a^2}+\frac{\left(x^2-c^2\right)^2}{c^4}=1$. Solving for $x$ yields three values: 0, which is the $x$-coordinate of the vertex of the parabola, and $\pm\frac{c}{a}\sqrt{2a^2-c^2}=\pm\frac{\sqrt{a^4-b^4}}{a}$

Because the y-coordinates of the two required intersection points are equal, the required distance between the two points is $\frac{2\sqrt{a^4-b^4}}{a}$.

In the given problem, $a=5$ and $b=\sqrt{7}$, so the needed distance is $\frac{2\sqrt{5^4-7^2}}{5}=\frac{48}{5}$.

The requested sum is $48+5=53$.

4. Hint: $f(x)f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right) \quad \Rightarrow f(x)=1\pm x^n$

$$\log_3\left(f(x)\right)=\log_3\left(4+\frac{4}{5}+\frac{4}{5^2}+\ldots\ldots\infty\right)\cdot\log_5\left(1+\frac{f(x)}{f\left(\frac{1}{x}\right)}\right)$$

$$\log_3\left(f(x)\right)=\log_3 5\cdot\log_5\left(1+\frac{f(x)}{f\left(\frac{1}{x}\right)}\right) \Rightarrow f(x)=1+\frac{f(x)}{f\left(\frac{1}{x}\right)}$$

$$\Rightarrow f(x)\cdot f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right) \quad \Rightarrow f(x)=x^2+1, \text{ as } \{f(10)=101\}$$

$$\therefore \sum_{r=1}^{n} f(r)=f(1)+f(2)+\ldots\ldots+f(x) \quad \because \sum_{r=1}^{6} f(r)=97<f(10)$$

5. Let $\alpha=p+iq, \beta=p-iq$

Since $\frac{\alpha^2}{\beta}\in R\Rightarrow q=\pm\sqrt{3}p \quad \therefore \left|\frac{2\alpha}{\beta}+1\right|^2=3$

6. Let $t=\cos x$. Then $\sec(2x)-\sec x=2$ becomes $\frac{1}{2t^2-1}-\frac{1}{t}=2,$

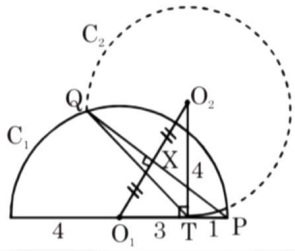


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which can be rewritten as $t=\left(2t^2-1\right)(2t+1)$. It follows that $4t^3+2t^2-3t-1=0$, which is equivalent to $(t+1)\left(4t^2-2t-1\right)=0$. Hence, $\cos x_1, \cos x_2$, and $\cos x_3$ are $-1, \frac{1+\sqrt{5}}{4}$, and $\frac{1-\sqrt{5}}{4}$, in some order. Therefore, $x_1, x_2$, and $x_3$ are $\pi, \frac{\pi}{5}$, and $\frac{3\pi}{5}$.

Then $\frac{m}{n}=\frac{\pi}{x_1+x_2+x_3}=\frac{1}{1+\frac{1}{5}+\frac{3}{5}}=\frac{5}{9}$. The requested sum is $10\cdot 5+9=59$.

7. We see that $O_2$ is reflection of $O_1$ over X, as $C_1, C_2$ have same radius and the same chord is equidistant from both centres.




Also, $O_2\ T=r=4\Rightarrow O_1O_2=5\Rightarrow O_2X=\frac{5}{4}$

$\Rightarrow PX^2=O_2P^2-O_2X^2=16-\frac{25}{4}=9.75 \quad \Rightarrow PQ^2=4PX^2=39$

8. Because we can factor $z$ from each vertex, the area of the triangle is equal to $|z|^2$ multiplied by the area of the triangle with vertices $2, 1+i, 1-i$.
The vertical side of that triangle has length 2 and the altitude to that side has length 1, so the area is 1.
We now just need the maximum possible value of $|z|^2$.
The equation $|4z-2|=1$ can be rewritten as $|z-1/2|=1/4$. Therefore, it exists as a circle with radius $1/4$ and center at $(1/2,0)$ on the complex plane. The maximum possible magnitude of a complex number on this circle occurs with $\frac{3}{4}$, with magnitude $\frac{3}{4}$. We can justify this because the line connecting the origin and the complex number $\frac{3}{4}$ goes through the center of the circle. Our answer is then $\left(\frac{3}{4}\right)^2=\frac{9}{16}$.

**9. Conceptual**

10. $A=(0,2,2)$

Vector alongside $AB=\begin{vmatrix} i & j & k \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix}=\hat{i}-\hat{j}$

Similarly vector along AC is $-\hat{i}+\hat{k}$ and AD is $-\hat{i}-\hat{j}-\hat{k}$.



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$\therefore$ Unit vector along AB, AD, AC are

$\frac{\hat{i}-\hat{j}}{\sqrt{2}}, \frac{\hat{i}-\hat{j}-\hat{k}}{\sqrt{3}}, \frac{-\hat{i}+\hat{k}}{\sqrt{2}}$ $\therefore \overline{AB} = \hat{i}-\hat{j}$ $\therefore B = (1,1,2)$

$\overline{AD} = \hat{i}-\hat{j}-\hat{k}$ $\therefore D = (1,1,1)$

$\overline{AC} = 2\left(\hat{i}-\hat{k}\right)$ $\therefore C = (2,2,0)$

$\therefore$ Volume of tetrahedron

$$= \frac{1}{6}\begin{vmatrix} 1 & -1 & 0 \\ 1 & -1 & -1 \\ 2 & 0 & -2 \end{vmatrix} = \frac{1}{3} \quad \overrightarrow{BC} = \hat{i}+\hat{j}-2\hat{k}, \overrightarrow{BD} = -\hat{k}$$

$$\overrightarrow{BC} \times \overrightarrow{BD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -2 \\ 0 & 0 & -1 \end{vmatrix} = -\hat{i}+\hat{j} \qquad \text{ar}(\triangle BCD) = \frac{1}{2}\left|\overrightarrow{BC} \times \overrightarrow{BD}\right| = \frac{1}{\sqrt{2}}$$

11. a) $f+g$ is discontinuous at 16 points

b) $f+g$ is continuous but non differential at $x = \frac{1}{5}$

12. Substitute $z+2i = x$

13. We have

$A^6 = B^6 = (ABC+2I)^2 = ABCABC + 4ABC + 4I$

$A\left(A^5 - BCABC - 4BC\right) = 4I$

So A is invertible. We have $0 \neq |\det A| = |\det B| = |\det C|$. So $B, C$ are invertible. Now

$B^5 = A^2BC^2 = A\left(B^3 - 2I\right)C = AB^3C - 2AC$

$= ABC^3 - 2AC = \left(B^3 - 2I\right)C^2 - 2AC = B^5 - 2C^2 - 2AC$

Since $C$ is invertible we have $C = -A$. So we have

$A^6 = B^6 = (2I - ABA)^2 = ABA^2BA - 4ABA + 4I$

$= AB^4A - 4ABA + 4I = A^6 - 4ABA + 4I$

$ABA = I$

$A^6 = B^6 = (2I - ABA)^2 = I^2 = I$

14. $AE \cdot AE = AB^2 = 9$ (Power of pt A)

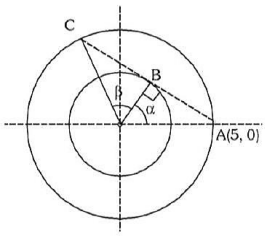
$AD \cdot AC = \frac{AB}{2} \times 2AB = AB^2$ $AE \cdot AF = AD \cdot AC \Rightarrow \frac{AE}{AD} = \frac{AC}{AF}$ $\triangle AED \sim \triangle ACF \Rightarrow \angle AED$



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$=\angle ACF$

BT




$B\equiv(4\cos\alpha,4\sin\alpha)$ $\quad B\equiv\left(\frac{16}{5},\frac{12}{5}\right)D\equiv\left(\frac{41}{10},\frac{6}{5}\right)M\equiv\left(\frac{55}{20},3\right)$

15. Let our subset be $\{11,12,13,\ldots,20\}$. If we add any element from the set $\{1,2,3,\ldots,10\}$ to our current subset, we will have to remove at least one element from our subset. Hence, the maximum size of our subset is $(C)$ 10.

16. We will solve this using states. Let $P_n$ be the probability of reaching 4, given that you start from $n$. We want to find $P_0$. Of course, $P_4=1$. We also know that

$$P_3=\frac{1}{4}P_4+\frac{1}{4}P_2$$

$$P_2=\frac{1}{4}P_3+\frac{1}{4}P_1$$

$$P_1=\frac{1}{4}P_2+\frac{1}{4}P_0$$

$$P_0=\frac{1}{2}P_1.$$

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Solving the system, we find that $P_1=\frac{2}{97}$ and $P_0=\frac{1}{97}$.

17. Let A′ and B′ be the feet of perpendiculars from the points A and B to the line of intersection of the two planes.

Equation of $AA'$ is $\frac{x-3}{2}=\frac{y+1}{-1}=\frac{z+1}{-1}=r$

for $A',2(2r+3)+(r+1)+(r+1)=5 \Rightarrow r=-\frac{1}{2}$

$\therefore$ Point A′ is $\left(2,-\frac{1}{2},-\frac{1}{2}\right)$.

In the same way, point $B'$ can be found as $(2,0,-1)$.

Let $A''$ be the point corresponding to $A$ when the plane $x+y+z=1$ is rotated through an angle $\frac{\pi}{2}$ about its line intersection with the plane $2x-y-z=5$.

The perimeter of $\triangle ABC$ would be minimum when A″, C, B are collinear. From the similar triangles $A''A'C$ and $BB'C$, we can observe that $C$ divides $A'B'$ in the ratio $P_1:P_2$ where $P_1,P_2$ are the perpendicular distances from $A,B$ to the planes $2x-y-z=5$ and $x+y+z=1$ respectively.



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$$P_1=\frac{|6+1+1-5|}{\sqrt{6}}=\frac{3}{\sqrt{6}}=\sqrt{\frac{3}{2}}$$

and $P_2=\frac{|3+1+0-1|}{\sqrt{3}}=\frac{3}{\sqrt{3}}=\sqrt{3}$

$\therefore$ point $C$ is $\left(\frac{2\sqrt{2}+2}{1+\sqrt{2}},\frac{-1}{\sqrt{2}\left(1+\sqrt{2}\right)},\frac{-1-\sqrt{2}}{\sqrt{2}\left(1+\sqrt{2}\right)}\right)=\left(2,\frac{-1}{\sqrt{2}\left(1+\sqrt{2}\right)},\frac{-1}{\sqrt{2}}\right)$

18. $x-f(x)=19\frac{x}{19}-90\frac{f(x)}{90}$

Plug in 1990,

$$1990-f(1990)=19\left[\frac{1990}{19}\right]-90\left[\frac{f(1990)}{90}\right]\Rightarrow\left[\frac{f(1990)}{90}\right]=\frac{f(1990)}{90}-\frac{14}{90}$$

$$\left\{\frac{f(1990)}{90}\right\}=\frac{14}{90}$$

Now,

$$\frac{1990}{90}<\frac{f(1990)}{90}<\frac{2000}{90}\Rightarrow\frac{f(1990)}{90}=21,22$$

Hence,

$$\frac{f(1990)}{90}=\frac{14}{90}+21,\frac{14}{90}+22\Rightarrow f(1990)=1904,1994$$

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# PHYSICS

19. The circuit becomes a Wheatstone bridge,
To minimise power in the $10\Omega$ resistor, current through it must be zero.
That happens when the bridge is balanced:

20. Energy in between $r=a$ to $r=2a$ gets converted to heat.

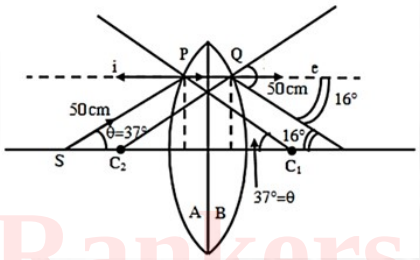
$$=\frac{q^2}{(2)4\pi\varepsilon_0\frac{(r_2r_1)}{(r_2-r_1)}}=\frac{q^2}{(2)4\pi\varepsilon_0(2a)} \qquad x=6$$

21. Speed of ball 2 after $1^{st}$ collision is $v_1$ $\qquad v_1=\frac{2(4)v_0}{(4+m)}$

Speed of ball 3 after $1^{st}$ collision with ball 2 is $v_2$ $\qquad v_2=\frac{2mv_1}{(m+9)}$

maximise $v_2 \Rightarrow m=6\ kg$

22.




By figure $i=\theta+\theta=2\theta=74^\circ$

by shell's law $\qquad 1\sin 74=\frac{8}{5}\sin\gamma \qquad 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right)=\frac{8}{5}\sin\gamma \quad \gamma=37^\circ$

So light becomes parallel to principle axis

hence $D=\infty \Rightarrow \frac{1}{D}=0$

23. $\frac{10-4}{2-0}=k\frac{4-3}{3-2}\Rightarrow k=3$

24. $\vec{M}=\frac{\pi I_0}{4}\left[a^2\hat{i}+b^2\hat{j}+c^2\hat{k}\right]$

$=\pi\left[9\hat{i}+b^2\hat{j}+c^2\hat{k}\right]$

$\vec{B}=(1.3)\frac{(-9\hat{i}-\hat{j}-4\hat{k})}{\sqrt{9^2+1^2+4^2}}$

$\vec{\tau}=\vec{M}\times\vec{B}=0$



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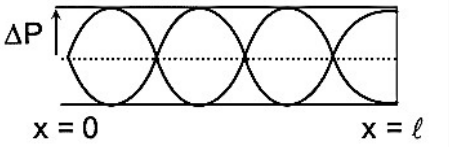
25. $d\phi = (2\pi x dx)kxt^2 \quad \phi = \frac{2}{3}kt^2\pi x^3 \qquad \frac{df}{dt} = \frac{4kptx^3}{3}$

$$E2\pi x = \frac{4\pi ktx^3}{3}; E = \frac{2}{3}ktx^2; \qquad d\tau = \left(\frac{2}{3}ktx^2\right)\frac{Q}{\pi R^2}(2\pi x dx)x$$

$$\int d\tau = \frac{4}{3}\frac{ktQ}{R^2}\int_0^R x^4 dx \Rightarrow \tau = \frac{4}{3}\frac{ktQ}{R^2}\frac{R^5}{5} \quad \text{At } t = 15\,\text{sec}, t = 5 \text{ N} - m$$

26. Use Nodal Analysis

27. Since the air column in the pipe vibrates in third overtone,




$$\ell = \frac{7v}{4f} = \frac{7\times 330}{4\times 660} = \frac{7}{8}\ m \quad \ell = \frac{7}{8}\ m$$

Now amplitude of pressure variation, $a = |\Delta P_0 \sin kx|$

$$a = \left|\Delta P_0 sin\left(\frac{7\pi x}{2\ell}\right)\right| \qquad \ldots(\text{ii})$$

$$\left(\because \frac{7\lambda}{4} = \ell \Rightarrow k = \frac{2\pi}{\lambda} = \frac{7\pi}{2\ell}\right)$$

At $x = \ell/3$, the amplitude of pressure variation is

$$a = \left|\Delta P_0 sin\left(\frac{7\pi}{2\ell}\times\frac{\ell}{3}\right)\right| - \left|\Delta P_0 sin\left(\frac{7\pi}{6}\right)\right| - \frac{\Delta P_0}{2}$$

The maximum pressure at $x = \ell 3$ will be $\left(P_o + \frac{\Delta P_0}{2}\right)$

At $x = \frac{\ell}{2}$, the amplitude of pressure variation is

$$a = \left|\Delta P_0 sin\left(\frac{7\pi}{2\ell}\times\frac{\ell}{2}\right)\right| = \left|\Delta P_0 sin\left(\frac{7\pi}{4}\right)\right| = \frac{\Delta P_0}{\sqrt{2}}$$

The minimum pressure at $x = \theta/2$ will be $\left(P_0 - \frac{\Delta P_0}{\sqrt{2}}\right)$

28. $Q = Wg + \Delta U \qquad n = \frac{W}{Q_{+ve}}$

29. $Q = CV = \frac{\varepsilon_0 A}{d} V \qquad \frac{dQ}{Q} = \frac{dA}{A} + \frac{d(d)}{d} + \frac{dV}{V} \simeq 0.045$

$$\frac{dE}{E} = \frac{dA}{A} + \frac{d(d)}{d} + \frac{2dV}{V} \simeq 0.055$$



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$F = \frac{Q^2}{2\varepsilon_0 A}$ $\frac{dF}{F} = \frac{dA}{A} + \frac{2d(d)}{d} + \frac{2dV}{V}$

When dielectric is inserted relative error remains same but absolute value increases.

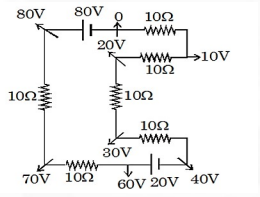
30. Induced emf in the loops, $\varepsilon = \frac{R^2\theta}{2}\frac{dB}{dt} = \frac{a_0 R^2\theta}{2}$

Induced current in the loop, $I = \frac{\varepsilon}{2R(1+\cos\theta)\lambda} = \frac{a_0 R\theta}{4\lambda\cdot(1+\cos\theta)}$

$V_O - V_Q = I\lambda 2R\cos\theta = \frac{a_0 R^2\theta\cos\theta}{2(1+\cos\theta)}$ At $\theta = 45^\circ$, $V_O - V_Q = \frac{a_0 R^2\pi}{8(\sqrt{2}+1)}$

$V_P - V_Q = e - I\lambda R = \frac{a_0 R^2\theta}{2} - \frac{a_0 R^2\theta}{4(1+\cos\theta)}$ $= \frac{a_0 R^2\theta}{4}\left(\frac{1+2\cos\theta}{1+\cos\theta}\right)$

For $\theta = 45^\circ$ $V_P - V_Q = \frac{a_0 R^2\pi}{8\sqrt{2}}$

31. 


$i = 1$ A $v_A - v_B = 40$ V $\theta_1 = 30$ $\theta_2 = 50$

32. Case-I

$mg - F = mR\alpha$

$FR = \frac{1}{2}MR^2\alpha$ $mg = \left(m + \frac{1}{2}M\right)R\alpha$ $R\alpha = \frac{mg}{m + \frac{1}{2}M}$

$F = \frac{1}{2}\frac{Mmg}{m + \frac{1}{2}M}$ $\Rightarrow \frac{Mmg}{m + \frac{M}{2}} = \frac{2Mg}{3}$

$3m = 2m + M$ $m = M$ $\alpha = \alpha' \Rightarrow \omega$ same

Case-2 $MgR = \frac{3}{2}MR^2\alpha'$

$Ra' = \frac{2g}{3}$ $Mg - F = MR'$ $F = \frac{Mg}{3}$

33. Let the length of rod submerged in the water be X.

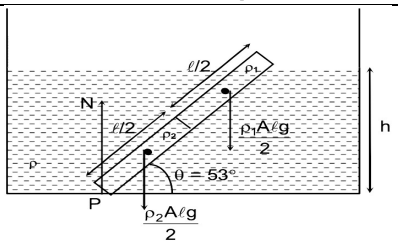
Since the rod is in equilibrium, $\tau_P = 0$

$\rho_1 \frac{A\ell g}{2}\frac{3\ell\cos\theta}{4} + \rho_2 \frac{A\ell g}{2}\frac{\ell\cos\theta}{4} = \rho Axg\frac{x}{2}\cos\theta$ $\frac{\ell^2}{4}(3\rho_1 + \rho_2) = \rho x^2$

$x = \frac{\ell}{2}\sqrt{\left(\frac{3\rho_1 + \rho_2}{\rho}\right)} = \frac{100}{2}\sqrt{\frac{1.50 + 2.50}{1}}$ $\Rightarrow x = 100$ cm



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Hence the depth of water in the tub, $h = x\sin\theta = 100\sin 53^\circ$

$h = 100 \times \frac{4}{5} = 80\ cm$ $\quad h = 80$ cm

34. In case 1, energy is lost in jerk when string suddenly becomes taut. But in case 2 there is no sudden jerk.

35. Given:

- Pitch $= 1$ mm
- Circular scale divisions $= 100$

Least count (LC):

$$\text{LC} = \frac{\text{Pitch}}{\text{No. of divisions}} = \frac{1}{100} = 0.01 \text{ mm}$$

Step 1: Find Zero Error

Given:

- Initially, circular scale lies between 2nd and 3rd main scale division
- After rotating $270^\circ$, the 3rd main scale division coincides with 12 th circular division

S Key Idea:

- $270^\circ$ rotation $= \frac{270}{360} \times 100 = 75$ divisions shift

So initial coincidence division: $12 - 75 = -63 \equiv 37 (\text{mod} 100)$

So initial reading $= 37$ divisions Zero Reading:

Main scale reading $(\text{MSR}) = 2$ mm

Circular scale reading (CSR $) = 37$ divisions

Observed reading $= 2 + 37 \times 0.01 = 2 + 0.37 = 2.37$ mm

Since nothing is placed, this is zero error.

Step 3: Correct Reading

Correct thickness = Observed – zero error $= 28.20 - 2.37 = 25.83$ mm

36. $F = mg - F_Z = mg - \frac{\pi R^2 P}{c}$ and from the law of conservation of energy, we can determine the sought velocity as $v = \sqrt{\frac{2FH}{m}} = \sqrt{2H\left(g - \frac{\pi R^2 P}{mc}\right)}$



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# CHEMISTRY

37. If 1.8 mole $CH_4$ contains 0.8 mole neutrons, the C in $CH_4$ has to be $C-14$ (instead of $C-12$)

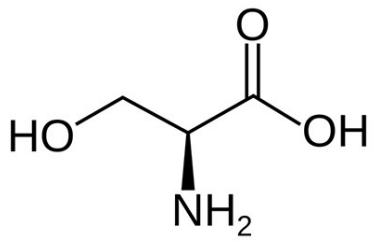
molar mass of $CH_4 = 18$ g $\quad n_{CH_4} = \frac{1.8}{18} = 0.1$ mole

moles of neutrons $= 0.1\times(14-6) = 0.8$

molar mass of $CH_2D_2 = 20$ g $\quad n_{CH_2D_2} = \frac{2}{20} = 0.1$ moles

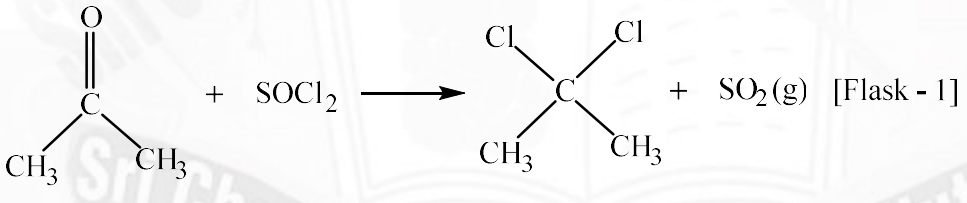
moles of protons $= 0.1\times(6+2+2) = \mathbf{1}$

38. The $\alpha$ – amino acid shown below is serine (one letter code **S**)




| | | | |
|---|---|---|---|
| *Sa* | ***Sb*** | ***Sc*** | *Sd* |
| ***Se*** | *Sf* | ***Sg*** | *Sh* |
| ***Si*** | *Sj* | *Sk* | *Sl* |
| ***Sm*** | ***Sn*** | *So* | *Sp* |
| *Sq* | ***Sr*** | *Ss* | *St* |
| *Su* | *Sv* | *Sw* | *Sx* |
| *Sy* | *Sz* | | |

39.




$Al_4C_3 + 12H_2O \longrightarrow 4\,Al(OH)_3 + 3\,CH_4(g)$ [Flask - 2]



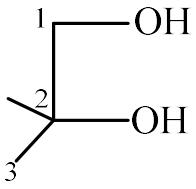

1 2

$$\frac{r_2}{r_1} = \frac{p_2A_2}{p_1A_1}\sqrt{\frac{M_1}{M_2}} = \frac{3}{1}\cdot\frac{A}{2A}\sqrt{\frac{64}{16}} = 3\times\frac{1}{2}\times 2 = \mathbf{3}$$



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40.




2-Methylpropane-1, 2-diol

$2+1+2=\mathbf{5}$

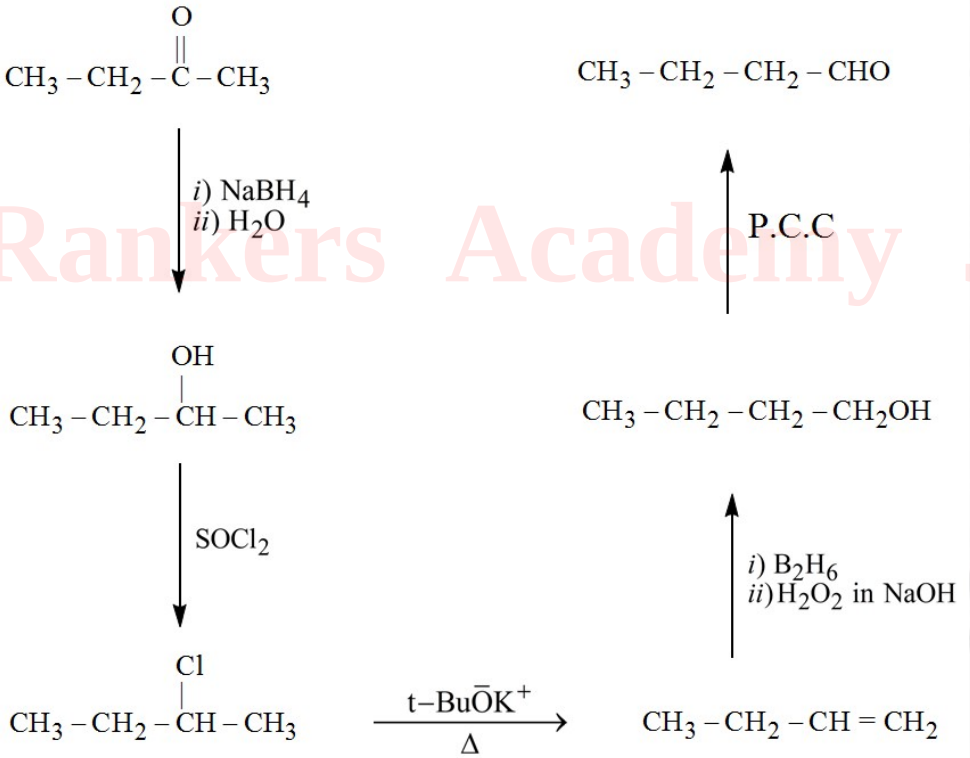
41. $pH = pK_a + \log\frac{[\text{Salt}]}{[\text{Acid}]}$ $4.5 = 4.199 + \log\frac{[\text{Salt}]}{[\text{Acid}]}$ $0.3010 = \log\frac{[\text{Salt}]}{[\text{Acid}]}$

$\log 2 = \log\frac{[\text{Salt}]}{[\text{Acid}]}$ $\therefore \frac{[\text{Salt}]}{[\text{Acid}]} = 2$

$\frac{0.2\times V}{0.1\times(300-V)} = 2, \quad V = 300-V, \quad 2V = 300, \quad V = 150 \text{ mL}$

Volume ratio $= \frac{V}{300-V} = \frac{150}{300-150} = \mathbf{1}$

42.




43. All 9 statements are correct (Refer p-block elements, Solutions & Biomolecules chapters of Class XII NCERT)

44. $F_2C = CF_2 + F_2 \longrightarrow F_3C - CF_3$ **(Y)**

$XeF_2 + C_2F_6 \xrightarrow{\text{Plasma}} F_3C - Xe - CF_3(\mathbf{Z}) + F_2$ (gaseous by-product)

$C_2H_6 + \underset{(\text{excess})}{F_2} \longrightarrow C_2F_6(\mathbf{Y})$



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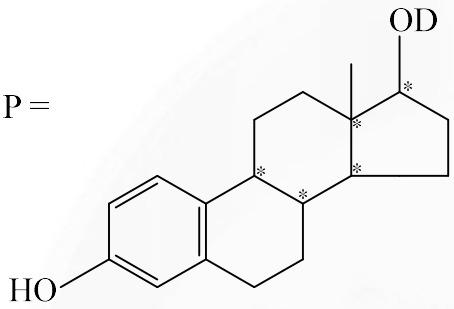
$a=3, b=4, c=21, d=6$

$\sqrt{(b+c)-(a+d)} = \sqrt{(4+21)-(3+6)} = \sqrt{25-9} = \mathbf{4}$

45.
- polar protic solvents (like 1-butanol) have a higher dielectric constant than polar aprotic solvents (like diethyl ether). The former therefore offers a more efficient separation of ions.
- In $S_{N_1}$ conditions involving chiral carbon centre, partial racemization takes place.

46.

P =




47. Refer Table 10.2, NCERT Class XI – s block elements.

| Property | Lithium Li | Sodium Na | Potassium K | Rubidium Rb | Caesium Cs |
|---|---|---|---|---|---|
| Atomic number | 3 | 11 | 19 | 37 | 55 |
| Atomic mass (g $mol^{-1}$) | 6.94 | 22.99 | 39.10 | 85.47 | 132.91 |
| Electronic configuration | [He] $2s^1$ | [Ne] $3s^1$ | [Ar] $4s^1$ | [Kr] $5s^1$ | [Xe] $6s^1$ |
| Ionization enthalpy / kJ $mol^{-1}$ | 520 | 496 | 419 | 403 | 376 |
| Hydration enthalpy/kJ $mol^{-1}$ | –506 | –406 | –330 | –310 | –276 |
| Metallic radius / pm | 152 | 186 | 227 | 248 | 265 |
| Ionic radius $M^+$ / pm | 76 | 102 | 138 | 152 | 167 |
| m.p. / K | 454 | 371 | 336 | 312 | 302 |
| b.p / K | 1615 | 1156 | 1032 | 961 | 944 |
| Density / g $cm^{-3}$ | 0.53 | 0.97 | 0.86 | 1.53 | 1.90 |
| Standard potentials $E^{\ominus}$/ V for ($M^+$ / M) | –3.04 | –2.714 | –2.925 | –2.930 | –2.927 |

48.
a) X is $Ma_3b_3$ which can exist in *fac* and *mer* forms, both the which contain a plane of symmetry (optically inactive)

b) *fac* – $[Co(dien)(NO_2)_3]$ contains a plane of symmetry passing through secondary nitrogen atom.

c) cation will exist in d & $\ell$ - forms, anion will exist in *fac* and *mer* – forms; hence a total of 4 stereoisomers will exist.

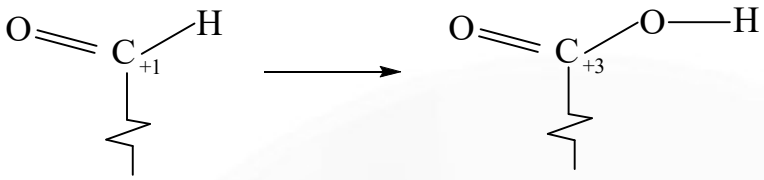
d) factual



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49. Salts like $NH_4Cl$ or $KNO_3$ that also contain covalent bonds in them (besides ionic bond) can also be used in salt bridge.

50. The adduct formation enables the reaction. The bonded fluorine atom acts as a Lewis base while $SbF_5$ acts as a Lewis acid.

51. 

Change in O.S. during oxidation $= 3-1=2$

$\therefore$ 2 mole electrons are released by 1 mole glucose

These 2 mole electrons must be accepted by $Br_2$ (3 moles)

$$3Br_2 + 2e^- \longrightarrow 2Br_3^-$$

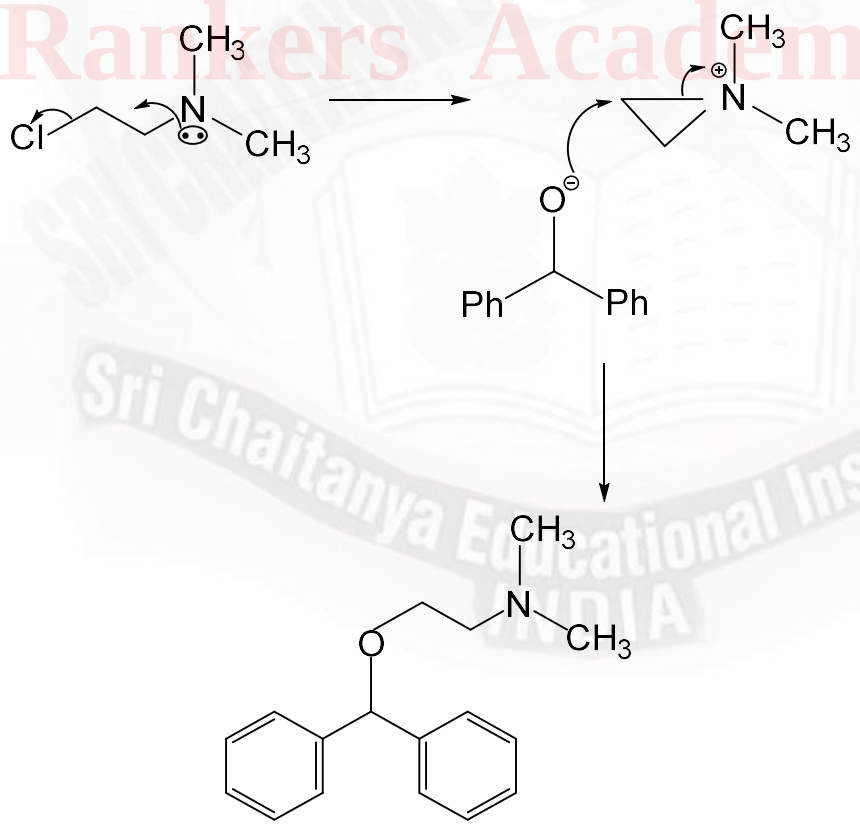
52. The reactant is base sensitive and can undergo $E_1cb$ reaction in the presence of hydroxide ions. Hence Wolff Kishner reduction is not suitable.

53. 1 mole Dolomite gives 2 mole $CO_2(g)$ (balanced reaction)

4 equivalent Dolomite gives 2 mole $CO_2(g)$

1 equivalent Dolomite gives $\frac{1\times 2}{4}=\frac{1}{2}$ mole $CO_2(g)$

$E.V_{CO_2}=11.2L$

54. 



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